

Case Study 1: Causal Effect of Fasting Blood Sugar on Heart Disease

A Causal AI Analysis using DoWhy

Dataset: Heart Disease UCI (Kaggle) | June 2026

1. Objective

This case study investigates whether elevated fasting blood sugar (FBS > 120 mg/dl) causally increases the risk of heart disease, or whether the observed correlation is an artefact of confounding variables such as age and cholesterol. Standard machine learning approaches treat this as a prediction problem; causal inference asks the harder question: what would happen if we intervened on FBS directly?

Variables

Treatment (T): Fasting Blood Sugar > 120 mg/dl (fbs = 1)

Outcome (Y): Presence of Heart Disease (target = 1)

Confounders (Z): Age, Sex, Cholesterol (chol), Resting Blood Pressure (trestbps)

Dataset size after preprocessing: 745 records, 6 features

2. Exploratory Analysis

Before applying causal methods, we examine the naive (observational) relationship between FBS and heart disease. This comparison does not account for confounding and should not be interpreted causally.

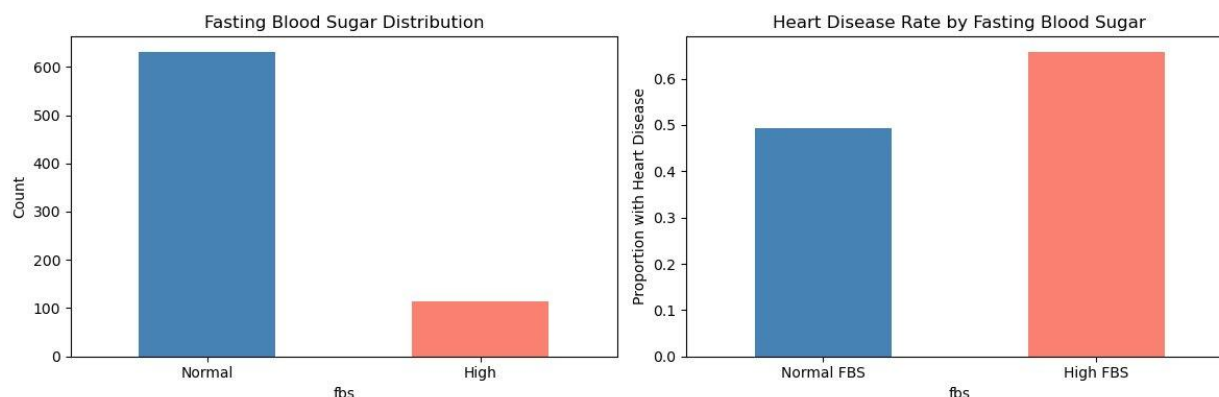


Figure 1: Fasting blood sugar distribution (left) and naive heart disease rate by FBS group (right)

Naive observation: The high FBS group shows a 66% heart disease rate compared to 49% in the normal FBS group — a raw difference of 17 percentage points. Taken at face value, this suggests high FBS is strongly associated with heart disease. However, the high FBS group is also slightly older and has marginally higher cholesterol, both of which independently increase heart disease risk. The causal analysis below disentangles these effects.

3. Causal Graph (DAG)

We encode our causal assumptions as a Directed Acyclic Graph (DAG). Each arrow represents a direct causal relationship. The confounders — age, sex, cholesterol, and resting blood pressure — point to both the treatment (fbs) and the outcome (target), creating backdoor paths that introduce spurious correlation.

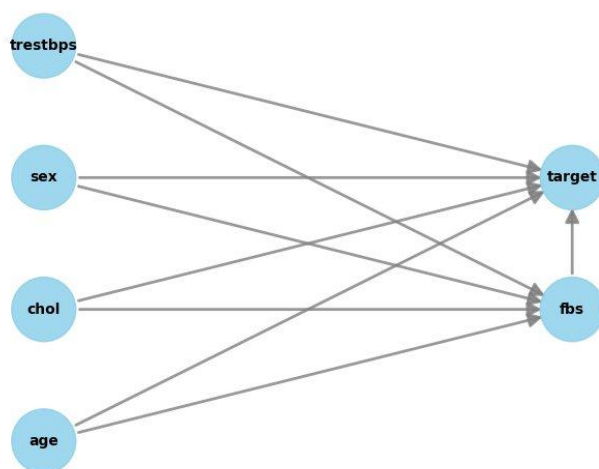


Figure 2: Causal DAG — confounders create backdoor paths between treatment and outcome

The backdoor criterion states that if we condition on a set of variables that blocks all backdoor paths from treatment to outcome, the remaining association is purely causal. DoWhy confirmed that conditioning on {age, sex, chol, trestbps} satisfies this criterion.

4. Causal Identification

DoWhy's identification step formally derives how to estimate the causal effect from observational data. It identified the backdoor estimand:

$$E[\text{target} \mid \text{do}(\text{fbs})] = d/d[\text{fbs}] \quad E[\text{target} \mid \text{age, chol, sex, trestbps}]$$

This expression means: the causal effect of FBS on heart disease can be estimated by taking the derivative of the conditional expectation of the outcome with respect to treatment, after adjusting for all confounders. This is the formal mathematical grounding for regression adjustment.

5. Causal Effect Estimation

The Average Treatment Effect (ATE) was estimated using linear regression adjustment on the backdoor variables. This method fits a regression of the outcome on treatment and all confounders, and reads off the coefficient on the treatment variable as the causal effect.

$$\text{ATE} = 0.043 \quad (\text{p-value: } 0.37)$$

$$\text{Naive correlation effect} = 0.165$$

The causal estimate of 0.043 is dramatically smaller than the naive correlation of 0.165. After adjusting for age, sex, cholesterol, and blood pressure, the true causal effect of high fasting blood sugar on heart disease is only ~4 percentage points — and is not statistically significant ($p = 0.37$). This is the central finding: the apparent 17% gap between groups was largely explained by confounders, not by FBS itself.

6. Refutation Tests

Two robustness checks were performed to validate the causal estimate:

Random Common Cause Test: A randomly generated confounder was added to the model. If the estimate is robust, it should barely change. Result: estimate moved from 0.04306 to 0.04314 — negligible change, confirming stability.

Placebo Treatment Test: The treatment variable was replaced with random noise. If the model is correctly specified, the causal effect should drop to near zero. Result: estimate collapsed to 0.004, confirming the original estimate was detecting a real (if small) signal and not noise.

7. Summary of Results

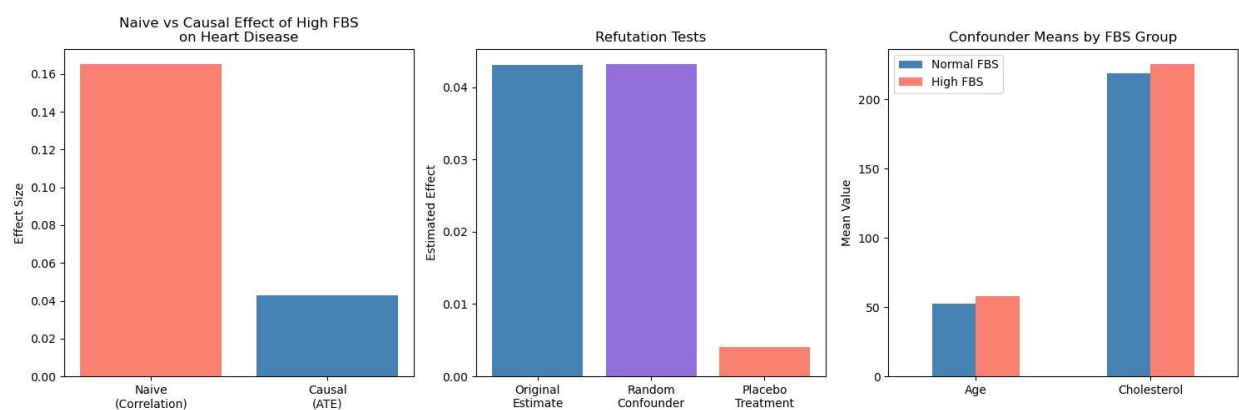


Figure 3: Naive vs causal effect (left), refutation test results (middle), confounder means by treatment group (right)

8. Conclusion

This case study demonstrates the core value of causal inference over naive correlation analysis. A purely observational approach would conclude that high fasting blood sugar significantly increases heart disease risk (17% difference). Causal analysis reveals this

effect is largely confounded — the true causal contribution of FBS is approximately 4% and statistically non-significant.

Key concepts demonstrated in this case study:

- Confounding and spurious correlation in observational data
- Structural Causal Models (SCMs) and DAG construction
- Backdoor criterion for causal identification
- Average Treatment Effect (ATE) estimation via regression adjustment
- Refutation testing for robustness validation

This result has direct clinical implications: interventions targeting FBS in isolation may be insufficient to reduce heart disease risk, as age and cholesterol are the stronger confounding drivers. A causal lens enables more targeted and effective intervention design.